

After Labour

Human Position in an AI-Robotic World System

From labour centrality to citizen claim, accumulated ownership, dependency, social reproduction, and Human Return

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Status and source boundary. This is a standalone K0 conceptual and measurement architecture, anchored in the prior standalone v1.0 and elevated into a fuller world-system model. It does not forecast a date for AGI, mass unemployment, or the disappearance of labour. Prior author papers are used as author-source genealogy: especially *Capitalism After Labor Centrality* (an undeposited internal working draft, not part of the deposited programme record; cited here for genealogy only), *Choice Begins Before Choice*, and *CTSA Human Return*. Their load-bearing objects are redefined in this manuscript so a reader need not consult them to reconstruct the argument. Relative to v1.0, this revision closes five missing loops: machine-wealth accumulation, scarce-asset rent burdens, aggregate-demand realization, social reproduction, and the separation of economic, informational, and coercive power. Self-citation establishes programme lineage, not independent validation. External literature supplies convergence, rivalry, empirical anchors, and defeaters. No original dataset is analyzed; unobserved parameters remain open. **Licence** CC BY 4.0. **Knowledge state** K0, version 1.2, timestamped and citable, not yet checked at an independent-reproduction tier above I1.

Edition note. This v1.2 is a source-lineage repair of the 6 September 2026 v1.1, made after an independent adversarial glosa review. Changes from v1.1: (i) the citation of *Capitalism After Labor Centrality* is now explicitly disclosed as an undeposited internal working draft at every mention, and its reference entry no longer uses a formal deposited-record citation format; (ii) the self-citations to *Choice Begins Before Choice* and *CTSA Human-Return Readout* now carry their deposited Zenodo DOIs; (iii) the self-citation to *Before Meaning, Before Choice* is corrected from the undeposited “v1.0” label to the actually deposited “Full Paper v1.1, glosa-checked” with its DOI; (iv) the reference to Bayraktar (2026) is corrected to the paper’s current canonical arXiv title, with the earlier circulated title noted; (v) one same/different comparison in §11 that used superiority language (“improves on”) and an untraceable “v0.1” label is rewritten to name the compared undeposited draft explicitly. No section, equation, table, proposition, or claim of the original v1.1 has been added, removed, renumbered, or strengthened. The v1.1 original remains the anchor record; this edition only repairs citation lineage and apparatus.

Central claim. The decisive transition in an AI-robotic world is not simply from human labour to machine labour. It is the weakening of labour as the universal bridge from personhood to a claim on social production. Once productive intelligence and embodied production can be reproduced and accumulated through capital at scale, a society must answer a different question: by what institutional route do humans retain claims on output, protection from scarce-asset rents, credible exit, live alternatives, corrigible agency, retained capability, and recognized social membership when the productive system needs less of their labour? A world can become richer while ownership compounds upward, demand weakens, care is underproduced, and the systemic position of most humans becomes weaker.

ABSTRACT

Artificial intelligence and robotics are commonly studied through productivity, task substitution, wages, labour share, market concentration, or inequality. This paper proposes a wider dependent variable: human systemic position. Anchored in the prior standalone model, it closes the world-system dynamics by adding five missing blocks. First, machine-productive wealth is allowed to accumulate and reproduce ownership asymmetry over time rather than treating broad ownership as an exogenous share. Second, scarce physical assets—especially housing, land, energy, and debt service—are separated from reproducible machine abundance, yielding an effective material claim after unavoidable rent burdens. Third, distribution feeds back into aggregate demand, realization, profits, and therefore the next round of machine-capital accumulation. Fourth, social reproduction is modeled as a distinct condition for maintaining human capability, care, and social continuity even when human labour becomes less necessary to production. Fifth, rule-setting

power is decomposed into economic, informational, and coercive channels rather than inferred directly from market concentration.

The model retains the v1.0 technology-production block, four-dimensional labour centrality, Citizen Claim Threshold, conversion-gate control, credible exit/dependency, live possibility, corrigible agency, Human Return, and social role. Human Systemic Position now depends on effective material claim after rent burdens, not gross claims alone. Two recursive loops close the model: an ownership-demand-production loop and a human-reproduction-political-rule loop. The paper therefore distinguishes productive necessity from social necessity: humans may become less necessary to scalable production while remaining indispensable to the reproduction of human life, legitimacy, care, and collective meaning. The central inequality of the AI age is not AI versus humans, but the growth of machine productive power and concentrated claims versus the growth of broad human ownership, effective material claim, exit, agency, retained capability, and social reproduction.

Keywords: artificial intelligence; robotics; political economy; labour centrality; ownership; transition control; dependency; citizen claim; live possibility; agency; Human Return; post-labour society.

1 THE QUESTION AFTER LABOUR

Industrial political economy linked mass participation to a practical chain:

human labour → production → wage → claim on output. (1)

Automation theory has long shown that technology can displace some tasks, complement others, and generate new tasks [1, 2]. The present paper does not assume that employment vanishes. Its problem begins when cognitive automation and embodied automation jointly make a growing share of productive capability reproducible through capital without a proportional increase in broad human labour.

The question is therefore not merely *how many jobs disappear*? It is whether the productive necessity of broad human labour can decline faster than the institutions that give people claims on output, bargaining power, usable alternatives, and standing. Automation can raise aggregate output while worsening distribution through wage and wealth channels [4, 5]. The remaining problem is wider: who owns productive intelligence, who controls the transitions through which it becomes action, who can route around those controls, what returns to humans as retained capability, and what replaces the social functions formerly organized by paid work?

2 STANDALONE READOUT DISCIPLINE AND AUTHOR-SOURCE GENEALOGY

2.1 Readout in this paper

This manuscript uses *readout* in a minimal, self-contained sense:

$$\text{Readout}_{Q,O,c}(S) = z, \quad z \neq S. \quad (2)$$

A readout is a finite, declared representation of a system relative to a question Q , reader or measurement interface O , and context c . The representation is not the whole system. Thus labour share, dependency, Human Return, and Human Systemic Position are readouts designed for declared questions; none is a substance inside a person or economy. This follows the programme's broader non-collapse discipline, but no root formalism is required to use the model here [36, 39].

2.2 What is inherited, and what is restated

The previous author paper *Capitalism After Labor Centrality* (an undeposited internal working draft, not part of the deposited programme record; cited here for genealogy only) developed four objects that are elevated here rather than treated as hidden premises: a four-dimensional labour-centrality profile, conversion-gate control, actor-specific credible exit/dependency, and a relational class-position vector [36]. *Choice Begins Before Choice* supplies the deeper

human distinction between formal feasibility and practically live alternatives [37]; CTSA supplies one operationalization of delayed unaided Human Return [38]. Every one of these objects is redefined below before it becomes load-bearing.

This matters methodologically. The paper can be understood after deleting its self-citations; the self-citations record genealogy and prior development, not evidential authority.

3 EMPIRICAL ANCHORS IN 2026

The model is motivated by current directional evidence, not calibrated from it. Table 1 records what the evidence does and does not license.

4 LITERATURE BOUNDARY AND RESIDUAL CONTRIBUTION

The paper does not claim that automation, inequality, platform power, substantive freedom, or human reliance on tools are new topics. Task-based economics already models displacement, complementarity, reinstatement, and labour-share effects [2, 3, 5]. Work on upstream AI competition shows why access prices and market power can affect downstream firms and workers [6]. Platform and intellectual-monopoly theories explain intermediation, network effects, knowledge exclusion, and rent [11–14]. Capability and non-domination traditions already reject the equation of formal resources with substantive freedom, while Foucault and Fricker show how power and interpretive resources can shape possible action before direct prohibition [7–10]. Human-AI research establishes that assisted performance, reliance, and later unaided performance can diverge [18–21].

The residual contribution is therefore architectural rather than a priority claim over these local mechanisms. Long-run wealth accumulation and wealth-income dynamics are already central to distributional economics [30], while housing is simultaneously a consumption necessity, a major household asset, and a source of distributional divergence [31]. Social-reproduction theory and care-economy research show that human production depends on background systems of care, households, education, and community reproduction that market production can underprovide [32, 33]. Recent work also links market concentration to political influence [34], and a 2026 working paper explicitly models a demand externality from automation when income shifts toward lower-propensity-to-consume owners [35]. The prior author paper *Capitalism After Labor Centrality* decomposed ownership, transition control, exit, dependency, claims, and relational class position [36]. The present paper integrates these neighboring concerns into one closed world-system architecture. Its strongest claim is that productive capacity, accumulated ownership, realized demand, scarce-asset rent, social reproduction, transition power, human agency, and political rule-setting must remain non-collapsed if the question is not merely “what happens to jobs?” but “what happens to the systemic position of humans when productive necessity, ownership, and agency can move in different directions?”

5 TECHNOLOGY BLOCK: CANDIDATE GENERATION, KNOWLEDGE, AND EMBODIMENT

Let G_t denote machine candidate-generation capacity: the ability to generate hypotheses, designs, code, plans, or other candidate outputs. Let Z_t denote validated knowledge available to the productive system. A minimal epistemic firewall is

$$\dot{Z}_t = v_t G_t - \delta_Z Z_t, \quad 0 \leq v_t \leq 1, \quad (3)$$

where v_t is a domain-specific conversion through testing, evidence, world resistance, or other validation. Hence

$$K_{\text{like}} \neq K_{\text{validated}}. \quad (4)$$

The model never infers knowledge growth directly from compute or benchmark growth.

Define operational AI capability separately as A_t^{AI} . For embodied automation, let

$$B_t^{RB} = N_t^{RB} q_t^{RB}, \quad (5)$$

where N_t^{RB} is deployed robot stock and q_t^{RB} effective per-unit capability. Then

$$\frac{\dot{B}_t^{RB}}{B_t^{RB}} = \frac{\dot{N}_t^{RB}}{N_t^{RB}} + \frac{\dot{q}_t^{RB}}{q_t^{RB}}. \quad (6)$$

IFR anchors the deployment term much better than the capability term.

A machine productive-capacity index is

$$M_t = (K_t^M)^\kappa (A_t^{AI})^\alpha (B_t^{RB})^\beta, \quad (7)$$

where K_t^M includes compute and machine-specific capital. Energy, materials, ecological constraints, trade restrictions, and geopolitical shocks enter a feasibility multiplier $\Xi_t \in (0, 1]$.

6 LABOUR CENTRALITY, NOT LABOUR DISAPPEARANCE

Employment is an incomplete indicator of labour's systemic role. Following the prior author-source construction but restating it here, define

$$L_t = \langle L_t^{\text{task}}, L_t^{\text{income}}, L_t^{\text{bottleneck}}, L_t^{\text{bargain}} \rangle. \quad (8)$$

The four coordinates ask, respectively: how much economically relevant task execution still requires humans; how important labour income is to household claims; how strongly production remains bottlenecked by non-substitutable human input; and how much leverage workers retain over terms, sequencing, adoption, and distribution [36].

Labour-Decentering Proposition [Open]. AI and robotics can reduce labour centrality without eliminating labour. Stable employment can coexist with falling bottleneck or bargaining centrality; high wages for some workers can coexist with a declining aggregate labour claim; and human-facing residual work can expand while scalable productive tasks migrate to machine systems.

Let $H_t = N_t^H h_t$ denote effective human productive capacity. A CES production block is

$$Y_t = \Xi_t [\omega_H H_t^\rho + \omega_M M_t^\rho]^{1/\rho}, \quad \rho = \frac{\sigma - 1}{\sigma}. \quad (9)$$

Under the restrictive benchmark of competitive factor pricing, the human labour-income share is

$$s_t^L = \frac{\omega_H H_t^\rho}{\omega_H H_t^\rho + \omega_M M_t^\rho}. \quad (10)$$

If $\sigma > 1$, sustained growth of M_t/H_t places downward pressure on s_t^L unless new human-intensive tasks, bargaining institutions, or complementary scarcity offset it. The paper does not assume that this holds economy-wide.

7 THE CLAIM CONSTITUTION: FROM WAGES TO CITIZEN CLAIMS

Suppose the broad population receives s_t^L of domestic output through labour income. Let $o_t \in [0, 1]$ be its beneficial ownership share of machine-capital income, including direct, pension, sovereign, cooperative, or social-fund ownership. Let $\tau_t \in [0, 1]$ be the effective redistribution rate applied to machine-capital income not already broadly owned. Define

$$q_t = o_t + \tau_t(1 - o_t) = 1 - (1 - o_t)(1 - \tau_t). \quad (11)$$

The broad material claim on domestic output is

$$\Gamma_t = s_t^L + q_t(1 - s_t^L). \quad (12)$$

This is a claim-accounting identity for the stylized model, not a welfare function. Public services can be added as a separate in-kind claim readout when defensibly valued; they are not silently equated with ownership.

For target claim $\bar{\Gamma} > s_t^L$, the minimum non-labour conversion required is

$$q_t^{\min} = \frac{\bar{\Gamma} - s_t^L}{1 - s_t^L}. \quad (13)$$

This is the Citizen Claim Threshold. As $s_t^L \rightarrow 0$, maintaining $\Gamma_t \geq \bar{\Gamma}$ requires $q_t \geq \bar{\Gamma}$. A society cannot preserve a labour-based distributive constitution merely by insisting that people keep working when labour ceases to be the main source of productive necessity.

For country i , define net cross-border machine-rent drain

$$D_{i,t}^{\text{rent}} = \frac{\text{Rent}_{i,t}^{\text{AI,out}} - \text{Rent}_{i,t}^{\text{AI,in}}}{Y_{i,t}}, \quad (14)$$

so that a simple domestic effective claim is

$$\Gamma_{i,t}^{\text{net}} = \Gamma_{i,t} - D_{i,t}^{\text{rent}}. \quad (15)$$

A country can automate locally while machine rents accrue abroad.

8 OWNERSHIP ACCUMULATION: THE STOCK THAT REPRODUCES THE NEXT CLAIM

The Citizen Claim Threshold in (13) is static unless the ownership share o_t is allowed to evolve. Yet machine-productive

Table 1: Current empirical anchors and interpretive constraints.

Anchor	Observed evidence	Constraint
AI scaling	Stanford AI Index 2026 reports global AI compute capacity growing about $3.3\times$ per year since 2022 and large one-year gains on some frontier benchmarks [23].	Compute growth is not knowledge growth; benchmark gain is not a universal capability growth rate.
AI cost	The AI Index reports dramatic task-dependent declines in inference cost over 2022–2025 [22].	Cheap access can accelerate deployment even without economy-wide AGI.
China robotics	IFR reports 295,000 industrial robots installed in China in 2024, 54% of global installations, and more than two million in operational stock; its 2026 release places AI-powered robotics at the center of national strategy [24, 25].	Installation and demand growth do not measure per-robot capability growth.
AI infrastructure	OECD identifies strong concentration in several AI-infrastructure layers, including hardware, fabrication, and cloud bottlenecks [26].	Concentration is layer-specific and is not identical to effective transition power.
Initial distribution	ILO reports a modest long-run decline in global labour income share, while World Inequality Report 2026 describes extreme wealth concentration [27, 29].	These conditions pre-date full AI-robotic substitution and cannot be attributed to AI alone.
Global asymmetry	ILO–World Bank analysis warns that some countries may experience disruption before capturing AI productivity dividends [28].	Adoption, local ownership, infrastructure, and cross-border rent capture must be separated.

assets generate returns that can be reinvested, inherited, pooled through pension or sovereign funds, diluted, taxed, or transferred. Long-run wealth-income evidence shows why the ownership stock cannot be treated as a passive background variable [30], while automation can amplify wealth inequality through differential exposure to capital returns [5].

Let $W_{i,t}^M$ denote actor i 's beneficial claim on machine-productive wealth. A minimal stock-flow equation is

$$W_{i,t+1}^M = (1 - \delta_W)W_{i,t}^M + r_t^M W_{i,t}^M + s_{i,t}^M + T_{i,t}^{\text{cap}} - \text{Tax}_{i,t}^{\text{cap}}, \quad (16)$$

where r_t^M is the realized return on machine-productive wealth, $s_{i,t}^M$ new saving or acquisition, $T_{i,t}^{\text{cap}}$ capital transfers or inheritance, and $\text{Tax}_{i,t}^{\text{cap}}$ taxes or dilution of the beneficial claim. The broad-population ownership share used in (11) is then

$$o_t = \frac{\sum_{i \in B} W_{i,t}^M}{\sum_i W_{i,t}^M}, \quad (17)$$

for a declared broad-population set B .

The point is not that $r^M > g$ or any other accumulation rule must hold universally. It is that ownership today is an input into ownership tomorrow. If machine rents are high and broad acquisition is weak, o_t can fall even while transfers keep current consumption stable. Conversely, social wealth funds, pensions, cooperatives, or public ownership can make o_t rise without dispersing operational control. Thus

$$\text{current redistribution} \neq \text{future ownership reproduction.} \quad (18)$$

9 SCARCE ASSETS, RENT BURDEN, AND EFFECTIVE MATERIAL CLAIM

Machine abundance is not abundance in every good. Housing, land, energy networks, water, urban location, spectrum, and some natural resources remain scarce or institutionally

rationed. OECD evidence shows that housing is both a major household asset and a major affordability burden, with prices and rents often rising faster than general inflation [31]. A post-labour society can therefore have very cheap reproducible digital and manufactured goods while households remain constrained by rents on scarce physical assets.

Define the household or population rent-burden share

$$B_{i,t}^{\text{scarce}} = \frac{R_{i,t}^{\text{house}} + R_{i,t}^{\text{land}} + R_{i,t}^{\text{energy}} + DS_{i,t}}{Y_{i,t}}, \quad (19)$$

where DS is required debt service. The effective material claim is

$$\Gamma_{i,t}^{\text{eff}} = \max\{0, \Gamma_{i,t}^{\text{net}} - B_{i,t}^{\text{scarce}}\}. \quad (20)$$

Gross transfers or dividends can therefore rise while effective command over discretionary life remains stagnant if housing, land, energy, or debt rents rise faster. This creates a second non-collapse:

$$\begin{aligned} &\text{machine abundance} \neq \text{low rent burden} \\ &\neq \text{effective material freedom.} \end{aligned} \quad (21)$$

10 AGGREGATE DEMAND AND REALIZATION: WHO CAN BUY THE AUTOMATED OUTPUT?

The production block describes technical output capacity. A market economy additionally requires realized demand. Let Γ_t^{eff} denote the distribution of effective household claims and m_t their marginal propensities to consume. Define aggregate demand

$$AD_t = C(\Gamma_t^{\text{eff}} Y_t, m_t) + I_t + G_t + NX_t, \quad (22)$$

and a simple realization ratio

$$\chi_t^{\text{dem}} = \min\left(1, \frac{AD_t}{Y_t}\right). \quad (23)$$

Realized machine-capital profit then depends on both productive capacity and demand realization,

$$\Pi_t^M = \chi_t^{\text{dem}} Y_t - \text{Cost}_t^M. \quad (24)$$

Recent work explicitly develops a demand externality of automation when income shifts toward owners with different consumption propensities [35]; the present paper uses only the weaker accounting implication that distribution can feed back into realization. Redistribution after labour is therefore not merely a welfare question. Under some parameterizations it can also become a condition for reproducing demand, revenue, and the accumulation process itself.

$$\Gamma^{\text{eff}} Y \rightarrow AD \rightarrow \Pi^M \rightarrow W_{t+1}^M \rightarrow M_{t+1}. \quad (25)$$

11 OWNERSHIP IS NOT TRANSITION POWER: CONVERSION GATES

Market concentration matters, but it is not yet a measure of what an actor can cause or block. Let e be a consequential transition – for example obtaining compute, changing provider, receiving payment, accessing a model, entering a market, or completing logistics. Let $V_t(e)$ be the best normalized viability of e under declared cost, time, quality, risk, and legal constraints, and $V_t(e | -j)$ the best viability after actor or gatekeeper j is removed. Define counterfactual conversion-gate control [36]:

$$G_{j,t}^{\text{conv}}(e) = \left[1 - \frac{V_t(e | -j)}{V_t(e)} \right]_+. \quad (26)$$

G^{conv} approaches zero when removing j leaves the transition essentially intact and approaches one when removal destroys viable routes.

For actor i , credible exit from j is

$$X_{i,t}(e; j) = \text{clip} \left(\frac{V_{i,t}(e | -j)}{V_{i,t}(e)}, 0, 1 \right). \quad (27)$$

Access and exit are therefore not identical. A person can have nominal access to a platform while having no credible substitute. An actor-specific dependency exposure for goal g is

$$D_{i \rightarrow j,t}(g) = \sum_{e \in E(g)} w_e(g) G_{j,t}^{\text{conv}}(e) [1 - X_{i,t}(e; j)]. \quad (28)$$

This differs from the concentration-only formulation in the undeposited draft *Capitalism After Labor Centrality* [36]: concentration is now treated as a possible upstream cause or proxy; the load-bearing human variable is dependency on consequential transitions with weak exit.

$$\text{Concentration} \neq G^{\text{conv}} \neq D. \quad (29)$$

If a concentrated provider is easily replaceable, transition power may be low. If many nominal competitors rely on one upstream bottleneck, firm count may understate transition power. Platform-capitalism and intellectual-monopoly theories strongly converge with the need to track intermediation and exclusion, but the present test is counterfactual rather than label-based [11–14].

12 RELATIONAL CLASS POSITION IN AN AI-ROBOTIC ECONOMY

The future social structure should not be read from income alone. For actor or household i , define a relational position

$$C_{i,t} = \langle O_{i,t}, G_{i,t}, \Gamma_{i,t}, A_{i,t}^{\text{access}}, X_{i,t}, D_{i,t}, R_{i,t}^{\text{rent}} \rangle, \quad (30)$$

where O is beneficial ownership, G gate control, Γ claim, A^{access} productive access, X credible exit, D dependency, and R^{rent} net rent position [36]. A stable class is not declared from one observation. It requires persistent clustering of these relations over time and reproduction through institutional rules.

The vector can represent familiar owner-worker relations when coordinates align, but also combinations characteristic of AI-era production: high-income professionals with little ownership and high infrastructural dependency; citizens with strong social dividends but little operational control; small firms with low ownership but high trusted routing rights; states with formal authority but dependence on private frontier expertise; or broad cooperative/public ownership with credible exit. These are candidate positions, not universal new classes.

13 THE POLITICAL-ECONOMY-TO-HUMAN BRIDGE

A political-economic relation cannot be upgraded directly into a claim about human agency. It must cross a named mechanism. Let

$$C_{i,t}^{PE \rightarrow H} = B_{PE \rightarrow H}(Z_t; i, g), \quad (31)$$

where the bridge can contain price, time, language, information, credit, permission, legal standing, identity, sanction, switching cost, network access, or institutional recognition. This is the point at which world-system structure changes the context in which options are experienced and acted upon.

13.1 Live possibility, defined here

For a person i and goal g , let

$$\Pi_{i,t}^{\text{live}}(g) \subseteq \Pi_{i,t}^{\text{feas}}(g) \subseteq \Pi_{i,t}^{\text{phys}}(g). \quad (32)$$

A live possibility is a feasible option that is sufficiently intelligible, discoverable, credible, normatively or socially usable, and actionable to enter practical consideration. It is not merely legally present. This definition is self-contained; the earlier author paper develops its history- and meaning-shaped microstructure in greater depth [37].

Let $\Lambda_{H,t}^{\text{live}} \in [0, 1]$ be a declared macro-readout of breadth and corrigibility of live alternatives. A candidate dynamic is

$$\begin{aligned} \dot{\Lambda}_{H,t}^{\text{live}} = & \lambda_1 B_t + \lambda_2 X_t + \lambda_3 P_t^{\text{plural}} \\ & + \lambda_4 r_t^H - \lambda_5 D_t - \lambda_6 C_t^I - \delta_\Lambda \Lambda_{H,t}^{\text{live}}, \end{aligned} \quad (33)$$

where B_t is material security and P_t^{plural} access to plural informational or interpretive routes. The equation is [Open]; it should be removed if ordinary affordability, salience, preference, or capability measures explain the same outcomes.

13.2 Corrigible human agency, defined here

For task g , define four conditions: R_g = can read relevant differences; D_g = acceptance or refusal remains attributable to the person; X_g = the decision can causally affect action; F_g = feedback, objection, correction, or revision remains possible. Current effective agency is

$$A_{H,t}^{\text{corr}}(g) = \max_{\pi \in \Pi_{i,t}^{\text{live}}(g)} \Pr^{\pi}(R_g \cap D_g \cap X_g \cap F_g). \quad (34)$$

This is a task-relative envelope, not a metaphysical free-will score. A larger formal option set does not guarantee larger agency; neither does mere action. The prior author work motivates this separation, but the present equation supplies the definition needed here [36, 37].

13.3 Human Return, defined here

AI-assisted system performance is not identical to human capability. Define a delayed unaided return profile

$$R_{H,t}^{\text{return}} = \langle C_t, T_t, S_t^{\text{skill}}, A_t^{\text{alt}} \rangle, \quad (35)$$

where the coordinates are Conceptual reconstruction, Tool selection, Skill execution, and Alternative generation after decisive AI assistance is removed. A normalized reporting readout $r_t^H \in [0, 1]$ may be constructed under frozen criteria. CTSA is one prior measurement architecture for these classes; the definition here does not require accepting CTSA as a validated scale [38]. External evidence already shows why assisted performance and later unaided capability must be separated [18–21].

14 SOCIAL ROLE AFTER LABOUR

Employment provides income but can also organize time, contact, status, activity, and collective purpose. Meta-analytic work on Jahoda's latent-deprivation tradition supports the relevance of these non-income functions [17]. Let $S_t^H \in [0, 1]$ denote recognized social role and standing. Let W_t be employment-based role provision and N_t non-work role infrastructure: care, family, religion, voluntary association, civic projects, learning, arts, sport, local production, or chosen work. A minimal dynamic is

$$\dot{S}_t^H = s_1 W_t + s_2 N_t - \delta_S S_t^H. \quad (36)$$

This does not claim that wage labour is psychologically necessary. It says that a society removing wage labour as a central institution must deliberately reproduce or replace functions that work previously bundled together. The time budget can shift as

$$1 = \ell^{\text{wage}} + \ell^{\text{care}} + \ell^{\text{learn}} + \ell^{\text{civic}} + \ell^{\text{leisure}}, \quad (37)$$

without a fall in ℓ^{wage} being automatically liberation or deprivation. Direction depends on voluntariness, claim, recognition, capability, and live alternatives.

15 SOCIAL REPRODUCTION: PRODUCTIVE NECESSITY IS NOT SOCIAL NECESSITY

A world-system theory is incomplete if it models humans only as labour inputs, consumers, or political actors. Human beings are born, raised, educated, healed, cared for, socialized, and supported through systems of social reproduction. Care-economy research documents the scale and gendered organization of paid and unpaid care across more than 90 countries [33]; Fraser's social-reproduction argument emphasizes that economic production depends on background capacities that capital accumulation can simultaneously strain [32].

Let H_t^{cap} denote a broad human-capability stock distinct from the productive input H_t in the CES block. A candidate reproduction dynamic is

$$\dot{H}_t^{\text{cap}} = f(\text{Care}_t, \text{Health}_t, \text{Education}_t, \text{Nutrition}_t, \text{Community}_t) - \delta_H H_t^{\text{cap}}. \quad (38)$$

This stock can affect future productive skill, agency, Human Return, social role, and the capacity to govern institutions, but it is not reducible to any one of them. Machine systems may assist care, diagnosis, education, and household work; they may also externalize or intensify unpaid coordination burdens. The sign is therefore open.

The central distinction is

$$\begin{aligned} &\text{productive necessity of humans} \\ &\neq \text{social necessity of human reproduction.} \end{aligned} \quad (39)$$

Humans can become less necessary to scalable production while remaining indispensable to the reproduction of human life, relationships, legitimacy, culture, and future generations. A post-labour settlement that preserves income but depletes care and social reproduction is not stable abundance; it is a transfer regime consuming its own human substrate.

16 THE HUMAN SYSTEMIC POSITION EQUATION

The central reporting object is not GDP. Let Γ_t^{eff} be effective material claim after cross-border and scarce-asset burdens, $A_{H,t}^{\text{corr}}$ corrigible agency, $\Lambda_{H,t}^{\text{live}}$ live possibility, r_t^H Human Return, S_t^H social standing, X_t^H credible exit, and D_t^H dependency. Define

$$P_t^H = \frac{(\Gamma_t^{\text{eff}})^{\theta_{\Gamma}} (A_{H,t}^{\text{corr}})^{\theta_A} (\Lambda_{H,t}^{\text{live}})^{\theta_{\Lambda}} (r_t^H)^{\theta_R} (S_t^H)^{\theta_S} (X_t^H)^{\theta_X}}{(1 + D_t^H)^{\theta_D}} \quad (40)$$

for declared non-negative reporting weights. This is not a welfare utility function and is not a validated cardinal scale. It is a typed audit index intended to prevent output, consumption, or transfer income from silently standing in for human position.

Taking log derivatives gives

$$\frac{\dot{P}_t^H}{P_t^H} = \theta_\Gamma \frac{\dot{\Gamma}_t^{\text{eff}}}{\Gamma_t^{\text{eff}}} + \theta_A \frac{\dot{A}_{H,t}^{\text{corr}}}{A_{H,t}^{\text{corr}}} + \theta_\Lambda \frac{\dot{\Lambda}_{H,t}^{\text{live}}}{\Lambda_{H,t}^{\text{live}}} + \theta_R \frac{\dot{r}_t^H}{r_t^H} \quad (41)$$

$$+ \theta_S \frac{\dot{S}_t^H}{S_t^H} + \theta_X \frac{\dot{X}_t^H}{X_t^H} - \theta_D \frac{\dot{D}_t^H}{1 + D_t^H}. \quad (42)$$

Hence the primary non-collapse result is

$$\dot{Y}_t > 0 \not\Rightarrow \dot{P}_t^H > 0. \quad (43)$$

A richer world can contain weaker humans.

17 ABUNDANCE WITHOUT AGENCY

The prior author paper introduced an Abundance Without Agency diagnostic; here it becomes a cross-check on (40). A candidate AWA state has

$$AWA_t = \langle Y \uparrow, \Gamma^{\text{eff}} \text{ adequate}, O_H \downarrow, X_H \downarrow, D_H \uparrow, \Lambda_H^{\text{live}} \downarrow, A_H^{\text{corr}} \downarrow, r_H \downarrow \rangle. \quad (44)$$

Material abundance can therefore coexist with structural dependence. This is not identical to poverty, unemployment, or authoritarianism. A population can receive cheap AI, public services, entertainment, health care, or transfers while owning little productive infrastructure, facing weak exit, or losing unaided capability in domains where AI becomes a permanent cognitive prosthesis [36].

18 RECURSIVE POLITICAL ECONOMY: THREE POWER CHANNELS

Technology is not an external shock to a passive society. Ownership, gate power, claims, dependency, and agency alter rule-setting; rules then alter the next ownership and gate structure. But economic concentration should not be treated as a complete theory of power. Empirical work finds that mergers and industrial concentration can be followed by increased lobbying activity, supporting a link between market structure and political influence without making the two identical [34].

Define a typed power vector

$$P_t = \langle P_t^{\text{econ}}, P_t^{\text{info}}, P_t^{\text{coerc}} \rangle. \quad (45)$$

P^{econ} concerns ownership, rents, credit, procurement, and conversion-gate control. P^{info} concerns control over search, ranking, language mediation, advertising, model access, data, and public attention. P^{coerc} concerns legally backed rule enforcement, taxation, policing, licensing, sanctions, and other state coercive capacities. These dimensions can align or diverge: a private platform may have high informational and economic power but little direct coercive authority; a state may hold coercive authority while depending on private compute or frontier expertise; public ownership can coexist with concentrated informational mediation.

Define broad political capacity

$$P_t^B \propto \Gamma_t^{\text{eff}} A_{H,t}^{\text{corr}} \Lambda_{H,t}^{\text{live}} X_t^H r_t^H, \quad (46)$$

and concentrated-system leverage as a function rather than a scalar identity,

$$P_t^E = P_E \left(P_t, \dot{G}^{\text{conv}}, D_t^H, 1 - \Gamma_t^{\text{eff}} \right). \quad (47)$$

Let I_t collect ownership, redistribution, exit, interoperability, competition, care/social-reproduction provision, and non-work institutional variables. It evolves as

$$\dot{I}_t = F_I(P_t^B, P_t^E, \text{state capacity, rules, shocks}). \quad (48)$$

No universal sign is imposed. Three recursive loops can coexist:

$$\begin{aligned} &\text{economic gate control} \rightarrow \text{rents} \rightarrow \text{political influence} \\ &\quad \rightarrow \text{future gate control,} \\ &\text{information control} \rightarrow \text{attention/interpretation} \\ &\quad \rightarrow \text{rule legitimacy} \rightarrow \text{future information control,} \\ &\text{productivity} \rightarrow \text{broad claim/exit} \rightarrow \text{agency} \\ &\quad \rightarrow \text{collective rules} \rightarrow \text{broader ownership/exit.} \end{aligned} \quad (49)$$

The future is institutionally path-dependent, not technologically predetermined. Crucially,

$$\text{ownership} \neq \text{informational power} \neq \text{coercive power}. \quad (50)$$

19 WORLD-SYSTEM REGIMES

The model yields diagnostics rather than deterministic forecasts. Table 2 distinguishes five analytically different orders.

A neo-feudal-like label should remain terminal and conditional, not a master concept. Concentration and rent are insufficient. A stronger diagnosis requires persistent hierarchical dependency, weak exit, rule asymmetry, and extra-economic or status-enforcing capacity. If those extra conditions are absent, the model should report rentier or concentrated capitalism rather than inflate the mode-of-production claim [14–16, 36].

20 WHY CHINA CHANGES TIMING BUT NOT LOGIC

China matters because it combines frontier AI research, the world's largest industrial robot deployment, large-scale manufacturing capacity, and a national strategy explicitly linking AI to physical robotics [23–25]. This can shorten the distance between cognitive automation and embodied production: candidate generation, design, planning, manufacturing, logistics, and machine iteration can increasingly occupy one industrial loop.

But the Chinese trajectory does not determine the world-system regime. State direction, public ownership, industrial policy, domestic manufacturer share, redistribution, household claims, and political rule-setting enter different coordinates than U.S. hyperscaler capitalism. The model therefore predicts neither one homogeneous AI capitalism nor one inevitable Chinese endpoint. It predicts competing automation blocs with different combinations of private ownership, state control, public claim, domestic value capture, external rent extraction, and exit.

Table 2: Five AI-robotic world-system regimes.

Regime	Production / claim	Control / agency	Likely human life pattern if stable
Augmented-labour capitalism	Human-machine complementarity and new tasks keep labour centrality substantial.	Moderate dependency; Human Return can rise if AI transfers skill.	Employment remains central but task composition changes; shorter hours become possible if productivity is shared.
AI rentier automation	Labour income centrality falls while $q < q^{\min}$; machine rents concentrate.	High gate power and dependency; weak exit and bargaining.	People remain consumers and service providers but increasingly rent access to productive intelligence and infrastructure.
Managed abundance	Labour centrality falls while transfers/public provision keep material claim high.	Control and dependency remain high; exit, pluralism, or Human Return may remain low.	Materially secure life with fewer work hours can coexist with managed dependence.
Shared automated abundance	Labour centrality falls while ownership/claims broaden and credible exit remains high.	Contestable gates, interoperability, plural routes, high Human Return.	Abundance widens effective options; work becomes more elective without reducing political standing.
Post-labour citizen economy	Wage labour is no longer the default basis of entitlement; citizen/social claims satisfy the threshold.	Agency, exit, retained capability, and rule contestability are institutionally protected.	Care, learning, religion, arts, civic activity, family, local production, and chosen work become primary recognized roles.

For countries outside the leading blocs, adoption alone is not success. The key questions are whether they own productive AI/robotic assets, capture domestic value, possess credible public or competitive alternatives, and convert machine capability into human capability rather than merely importing rented intelligence. This is the world-system form of the ILO–World Bank warning that disruption can precede dividend [28].

21 WHAT HUMANS MAY ACTUALLY DO

The model does not predict mass idleness. It predicts a possible separation between economic necessity and human activity. Even in highly automated systems, societies may continue to demand human responsibility, embodiment, trust, legitimacy, or presence in care, relationships, law, politics, religion, culture, science direction, art, status competition, local community, luxury human service, sport, and institutional accountability.

The decisive distinction is whether these become chosen and socially recognized roles or low-bargaining residual tasks left after machine substitution. A humane post-labour order therefore requires more than income replacement. It requires a new social constitution of contribution in which people can matter without proving their right to subsistence through scarce wage labour.

The institutional transition is

$$\text{labour-based claim} \longrightarrow \text{human/citizen claim} \quad (51)$$

on social production.

Candidate mechanisms include broad pension and sovereign ownership, social wealth funds, cooperative AI, public compute, universal basic services, social dividends, taxation of machine rents, and competition/interoperability rules. They are not interchangeable: redistribution can raise Γ while leaving ownership, gate power, or exit unchanged.

22 MEASUREMENT AND FALSIFICATION

The architecture earns scientific value only if its additional objects discriminate outcomes beyond standard productivity, wage, task-exposure, and market-structure measures.

No single dataset can identify the full river. Ownership and claims require balance sheets, pension/fund exposure, corporate and public ownership records, inheritance and capital-flow data. Scarce-asset burdens require housing, land, energy, and debt-service accounts. Demand realization requires distributional consumption and propensity-to-consume data. Social reproduction requires care-time, health, education, household, and community indicators. Gate control requires contracts, technical dependency graphs, outages, API/cloud pricing, procurement, standards dependencies, and substitution tests. Exit requires switching time, portability, retraining cost, legal interoperability, and post-switch quality. Human Return requires delayed unaided tasks rather than only contemporaneous productivity. Where causal identification is unavailable, the framework should report a descriptive transition map rather than upgrade correlation into domination. Table 3 lists the falsification-oriented programme.

23 WHAT CAN AND CANNOT BE FORECAST NOW

Three quantities remain too weakly identified for a defensible year forecast: economy-wide substitution elasticity between humans and AI-robot capacity; annual growth of effective per-robot capability q_t^{RB} ; and the institutional response path of ownership, redistribution, exit, and Human Return. Therefore the paper does not calculate a date at which humans “leave the economy.”

A normalized early-warning diagnostic can instead com-

Table 3: Falsification-oriented programme.

Test	Prediction if the proposed layer matters	Removal / compression condition
Gate vs concentration	Counterfactual removal/substitution predicts transition failure beyond HHI or top-share measures.	Remove G^{conv} if concentration/interoperability fully explain outcomes.
Exit vs access	Equal nominal access but different credible exit predicts price exposure, bargaining, refusal, or switching.	Collapse X into access if it adds no held-out value.
Labour profile	Task, income, bottleneck, and bargaining centrality dissociate across sectors or time.	Replace L with simpler employment/labour-share measures if no stable dissociation appears.
Citizen Claim Threshold	Falling s^L predicts weaker broad command over output unless q rises at least as fast as (13).	Revise if observed claims are better captured by another accounting decomposition.
Live possibility	Bridge mechanisms alter considered alternatives or switching after formal rights/resources are controlled.	Compress into affordability, capability, salience, or preference if no incremental value remains.
Human Return	Similar AI-assisted productivity can coexist with different delayed unaided concept/tool/skill/alternative performance.	Remove r^H if delayed independent measures add no information beyond ordinary skill measures.
AWA	High material provision can coexist with low ownership, exit, agency, or Human Return.	Reject AWA as distinct if material welfare tightly tracks those dimensions.
Social-role substitution	Reduced employment is less damaging where non-work institutions replace time structure, status, social contact, purpose, and activity.	Redesign S^H if these mediators do not explain outcomes.
Ownership accumulation	Persistent differences in machine-wealth returns, acquisition, inheritance, and taxation predict the evolution of broad ownership o_t beyond current income flows.	Treat o_t as exogenous if stock-flow ownership dynamics add no predictive value.
Scarce-asset burden	Similar gross citizen claims can yield different discretionary command where housing, land, energy, or debt burdens differ.	Remove B^{scarce} if gross and effective claims are empirically indistinguishable.
Demand realization	Distribution of effective claims and consumption propensities predicts revenue/realization after controlling productive capacity.	Drop χ^{dem} if demand never constrains realized output/profits in the studied regime.
Social reproduction	Care, health, education, nutrition, and community capacity predict future human capability and institutional resilience beyond wage employment alone.	Compress H^{cap} if these variables add no stable explanatory value.
Power channels	Economic, informational, and coercive power dissociate across actors and predict different transition outcomes.	Collapse P if one scalar gate/dependency measure dominates across the declared domain.

pare decoupling pressures:

$$\begin{aligned}\Omega_t = & w_M(g_M - g_H) + w_D g_D + w_G g_G \\ & - w_q g_q - w_\Gamma g_{\Gamma^{\text{eff}}} - w_X g_X \\ & - w_R g_{r^H} - w_\Lambda g_\Lambda - w_H g_{H^{\text{cap}}}.\end{aligned}\quad (52)$$

The first line groups productive and dependency pressure; the remaining terms are institutional and human counter-response. All terms require normalization before comparison. Persistent $\Omega_t > 0$ is an early-warning condition, not a deterministic forecast.

24 CLAIM BOUNDARIES

The theory should be narrowed rather than protected when its extra variables fail. Six boundaries are especially important.

- Machine candidate generation is not validated knowledge; (3) is a bookkeeping firewall, not a law of science.
- Robot installations are not robot capability; the second term in (6) remains open.
- The CES block is a scenario architecture, not a claim that one substitution elasticity governs the world economy.

- The Citizen Claim Threshold is an accounting result inside the stylized claim block; it does not choose between ownership, taxation, services, dividends, or other institutions.
- The wealth-accumulation and aggregate-demand blocks are stock-flow and realization architectures, not universal laws of returns, saving, or consumption. Their parameters must be estimated by jurisdiction and period.
- Effective material claim depends on declared scarce-asset burdens; it must not be used to imply that all housing, land, energy, or debt payments are exploitative rents.
- Human Systemic Position is an audit index, not a universal welfare score. Its weights must be declared and sensitivity-tested.
- Concentration, dependency, live-field narrowing, or managed abundance do not by themselves establish domination, structural violence, or neo-feudalism. Those stronger claims require mechanisms, feasible counterfactuals, restricted objection/exit, persistence, and independent normative justification.

A strong story of human marginalization would be substantially weakened if future evidence shows concentrated AI ownership together with low switching costs, robust interop-

erability, broad citizen claims, strong bargaining, credible public alternatives, contestable rule-setting, and stable or rising Human Return. Under that pattern,

$$\text{concentration} \not\Rightarrow \text{dependency} \not\Rightarrow \text{agency loss.} \quad (53)$$

The correct description might then be ordinary capitalism with unusually productive technology, or shared abundance with concentrated production but strong countervailing institutions.

25 CONCLUSION: FROM PRODUCTIVE NECESSITY TO HUMAN MEMBERSHIP

The central economic question of advanced AI is often posed as “what happens to jobs?” The question remains vital, but it begins too late. If productive cognitive and embodied capability becomes reproducible through capital, labour can lose task, income, bottleneck, or bargaining centrality while employment remains positive. The social contract then faces a constitutional problem: the old bridge from labour to wages to membership no longer guarantees broad claims on the abundance the system can produce.

The standalone river developed here is

$$\begin{aligned} &\text{candidate generation} \rightarrow \text{validated } K/A^{AI} \\ &\rightarrow \text{robotic embodiment} \rightarrow M_t \rightarrow Y_t \\ &\rightarrow L_t \rightarrow s_t^L \rightarrow \Gamma_t \\ &\rightarrow W_t^M \rightarrow o_{t+1} \rightarrow \Gamma_t^{\text{eff}} \\ &\rightarrow AD_t \rightarrow \Pi_t^M \rightarrow W_{t+1}^M \\ &\rightarrow \{G^{\text{conv}}, X, D\} \rightarrow \text{human bridge} \\ &\rightarrow \{\Pi^{\text{live}}, A_H^{\text{corr}}, r_H, S_H\} \\ &\rightarrow H_t^{\text{cap}} \rightarrow P_t^H \rightarrow P_t \\ &\rightarrow \text{collective rules} \rightarrow \text{next system state.} \end{aligned} \quad (54)$$

The river is now closed by two feedback systems. The ownership-demand-production loop is

$$\Gamma^{\text{eff}} Y \rightarrow AD \rightarrow \Pi^M \rightarrow W_{t+1}^M \rightarrow \{o_{t+1}, M_{t+1}\}, \quad (55)$$

while the human-reproduction-rule loop is

$$\begin{aligned} &\{\text{Care, Health, Education}\} \rightarrow H^{\text{cap}} \rightarrow \{A_H^{\text{corr}}, r_H, S_H\} \\ &\rightarrow P \rightarrow I_{t+1}. \end{aligned} \quad (56)$$

A world-system regime is stable only if both the production-claim circuit and the human-reproduction-governance circuit reproduce themselves.

The critical inequality is therefore not AI versus humans. It is

$$\begin{aligned} &\text{growth of machine productive power} \\ &\text{versus growth of broad human claim, exit,} \\ &\text{agency, and Human Return.} \end{aligned} \quad (57)$$

If the first persistently outruns the second, a richer world can contain weaker humans. If the second keeps pace, the same technology supports the opposite historical possibility: humans cease having to sell most of their waking lives merely to justify a claim on what their civilization can produce.

AUTHOR DECLARATIONS

Research status. Conceptual and measurement-architecture paper; full world-system standalone revision anchored in v1.0; no original empirical dataset is analyzed.

Author-source use. Prior programme papers are cited as conceptual genealogy and prior formal development. This manuscript restates every load-bearing object required for standalone comprehension.

AI assistance disclosure. Generative AI assisted retrieval, adversarial review, formalization, drafting, and type-setting. The human author selected the research problem, programme lineage, substantive theoretical commitments, and retains responsibility for claims, citations, and revisions.

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